

Relational Architecture of Emergent Dimensionality via Reduction of Relational Radius R : Conduction, Optical Guiding, and Magnetism as Constraints on Degrees of Freedom

A strict exploratory framework with pre-registered falsifiers

Luis Diego Mata Sánchez

Abstract

We propose a strict exploratory framework in which observable material properties are interpreted as *architectures of constraint* on degrees of freedom that act as bridges reducing a system’s *relational radius* R of information. Distance is treated operationally through phase mismatch relative to a universal reference c and a class-fixed scale ω^* , rather than as a pre-existing container. The manuscript is organized with explicit separation of levels: postulates, internal definitions/results, tentative physical correspondences, falsifiable predictions, and programmatic speculation. A minimal effective-field implementation is given for 2D phase coherence (XY/BKT) with a single independently estimated control knob κ encoding activation of out-of-plane dephasing channels. To avoid the critique “ κ only renames J ”, we pre-register two minimal non-degenerate extensions (vortex-core energetics and next-nearest neighbor coupling) and specify BKT diagnostics (collapse, universal jump, $\eta = 1/4$) together with a disciplined transport protocol against null models. The framework is designed to be directly testable against existing datasets (open when available, otherwise digitizable or obtainable by request), with pre-registered success/failure criteria.

Contents

1	Level separation (scope and discipline)	2
2	Minimal operational dictionary	3
2.1	Relational radius from phase mismatch	3
2.2	Static vs dynamic structure: β and Γ	3
2.3	How to fix ω^* (pre-registered by class)	3
3	Relational interpretation of material “properties” (tentative correspondences)	3
3.1	Conductor: potential gradient as a restriction condition (not “energy as primitive”) . .	3
3.2	Optical fiber: guided modes as controlled growth of $ \Delta\phi $	4
3.3	Ferromagnet: collective rotational mode as reduced relational fluctuations	4
3.4	Ontological sieve analogy	4
4	2D coherence model (operational core)	4
4.1	Base effective model (XY on a graph)	4
4.2	Control knob κ (independently estimated)	4
4.3	Pre-registered non-degenerate extensions (to block “ κ just rescales J ”)	4

4.4	Temperature and dynamics (Langevin form)	5
4.5	Observables	5
4.6	Helicity modulus (BKT diagnostic)	5
5	Transport ansatz (phenomenology with guardrails)	5
5.1	Pre-registration discipline (pass/fail guardrails)	6
5.2	Null models (required comparison)	6
6	Falsifiable predictions (kill tests)	6
6.1	Collapse variable	6
7	Experimental mapping (data-first strategy)	7
7.1	Independent κ assignment (examples)	7
7.2	Priority dataset that is explicitly open (optical phase noise)	7
7.3	Why graphene remains a natural first target	7
8	Limitations (explicit)	7
9	Quarantined programmatic layer (non-essential)	7
A	Workbench questions	7

1 Level separation (scope and discipline)

To prevent category mistakes (mixing ontology, effective modeling, and validation), we separate five layers:

0.1 Postulates

Explicit assumptions required only to define an operational dictionary.

0.2 Internal constructs and algebraic results

Definitions and model equations (or minimal modeling choices) that follow from the postulates.

0.3 Tentative physical correspondences

Interpretive mappings that *do not* count as validation.

0.4 Falsifiable predictions

Quantitative tests with pre-registered pass/fail criteria.

0.5 Programmatic speculation

Quarantined ideas not used to justify any falsifier.

2 Minimal operational dictionary

2.1 Relational radius from phase mismatch

Definition 1 (Relational radius). *For two subsystems indexed by i, j , define an effective relational radius*

$$R_{ij} := \frac{c}{\omega^*} |\Delta\phi_{ij}|, \quad \Delta\phi_{ij} := \phi_i - \phi_j, \quad (1)$$

where ω^* is a class-fixed reference frequency (fixed by experimental class, never by sample), and c is the universal velocity reference.

Operational use via proxies. In many platforms $|\Delta\phi|$ is not directly observed; one measures phase variance or coherence lengths. Therefore Eq. (1) is used through proxies:

$$R \propto \sqrt{\langle(\Delta\phi)^2\rangle}, \quad R \propto \ell_\phi^{-1} \text{ (electronic transport via WL/WAL dephasing fits)}. \quad (2)$$

This is an explicit operational mapping (not a microscopic derivation).

2.2 Static vs dynamic structure: β and Γ

Definition 2 (Static vs dynamic material operators). *We separate two roles that materials can play:*

1. **Static constraint** β : *restrictions on accessible degrees of freedom in equilibrium descriptions (geometry/topology of allowed modes).*
2. **Dynamic coupling** Γ : *coupling to dephasing channels/baths controlling phase-noise injection and relaxation rates.*

In equilibrium contexts, Γ^{-1} often covaries with β heuristically, but no universal dependence is assumed.

2.3 How to fix ω^* (pre-registered by class)

We pre-register ω^* by experimental class:

- **Electronic transport:** $\omega^* \approx v_F/\ell_0$, with fixed reference length ℓ_0 (e.g. device length L). Example: $L = 1 \mu\text{m}$, $v_F \approx 1 \times 10^6 \text{ m/s}$ gives $\omega^* \sim 1 \times 10^{12} \text{ s}^{-1}$.
- **Optics / fibers:** ω^* is the injected carrier frequency (or stated bandwidth). For the phase-noise dataset associated with Ref. [1], a natural choice is telecom C-band: $\omega^* = 2\pi \times 193 \text{ THz}$.
- **Magnonics:** ω^* is the excitation frequency of the driven spin-wave mode.

No per-sample tuning of ω^* is permitted.

3 Relational interpretation of material “properties” (tentative correspondences)

3.1 Conductor: potential gradient as a restriction condition (not “energy as primitive”)

A conductor is interpreted as a structure whose geometry and microstate constraints reduce phase dispersion among charge carriers, i.e. it reduces internal R relative to the environment.

Statement (non-claim). A potential gradient is treated not as a primitive “energy” object but as a boundary condition that channels and aligns electronic states, reducing effective dephasing pathways (slower growth of $|\Delta\phi|$ with distance).

3.2 Optical fiber: guided modes as controlled growth of $|\Delta\phi|$

A fiber is a phase-mismatch regulator: modal structure restricts transverse degrees of freedom and suppresses coupling to radiation modes, reducing the growth of $\langle(\Delta\phi)^2\rangle$ with propagation distance.

3.3 Ferromagnet: collective rotational mode as reduced relational fluctuations

Magnetism is interpreted as a collective alignment mode that reduces effective relational fluctuations for a subset of degrees of freedom, lowering internal R compared to disordered phases.

3.4 Ontological sieve analogy

A structured medium acts as an *ontological sieve*: it filters modes by compatibility with its allowed degrees of freedom. “Hole smaller than the wave” is read as “no compatible channel exists for that mode”, so reflection/attenuation/guiding emerges from constraint architecture.

4 2D coherence model (operational core)

4.1 Base effective model (XY on a graph)

Let $G = (V, E)$ be a lattice/graph with $|V| = N$ sites and $|E|$ bonds. Assign a phase variable $\phi_i(t)$ to each site:

$$H_{\text{eff}0} = - \sum_{\langle i,j \rangle \in E} J_{ij} \cos(\phi_i - \phi_j) + \sum_{i \in V} V_i(\phi_i). \quad (3)$$

V_i can be set to zero initially; optionally a weak random term models disorder.

4.2 Control knob κ (independently estimated)

Definition 3 (κ as channel activation index). $\kappa \in [0, 1]$ encodes activation of out-of-plane (or extra) dephasing channel length extractions $[\beta - 5]$.

4.3 Pre-registered non-degenerate extensions (to block “ κ just rescales J ”)

Baseline stiffness renormalization (allowed but not sufficient alone):

$$J_1(\kappa) = J_0(1 - \kappa). \quad (4)$$

Extension A (vortex-core energetics).

$$E_c(\kappa) = E_{c0}(1 - \lambda \kappa), \quad (5)$$

with *global* λ (pre-registered, not per sample). This changes vortex fugacity and finite-size flow beyond pure stiffness rescaling.

Extension C (next-nearest neighbor coupling).

$$H_{\text{eff}} = -J_1(\kappa) \sum_{\langle i,j \rangle} \cos \Delta\phi_{ij} - J_2(\kappa) \sum_{\langle\langle i,j \rangle\rangle} \cos \Delta\phi_{ij} + \sum_i V_i(\phi_i), \quad J_2(\kappa) = \gamma \kappa J_0, \quad (6)$$

with *global* γ (pre-registered). This breaks the degeneracy “ κ is just J ” in a controlled way.

4.4 Temperature and dynamics (Langevin form)

When transport-like time scales matter:

$$\dot{\phi}_i = -\Gamma(\kappa) \frac{\partial H_{\text{eff}}}{\partial \phi_i} + \eta_i(t), \quad \langle \eta_i(t) \eta_j(t') \rangle = 2\Gamma(\kappa) T \delta_{ij} \delta(t - t'). \quad (7)$$

Here T is an effective temperature in model energy units (Boltzmann constant absorbed). Physical temperature enters via calibration (e.g. $T_{\text{phys}} = (J_0/k_B) T_{\text{model}}$ after choosing J_0). $\Gamma(\kappa)$ is a phenomenological damping rate that may increase with κ (stronger coupling to external channels).

Method separation (recommended). Use Monte Carlo for equilibrium BKT diagnostics; use Langevin for dynamics/relaxation/transport-like signatures. This prevents mixing equilibrium and nonequilibrium claims.

4.5 Observables

- **Order parameter:** $\Psi := \left| \frac{1}{N} \sum_{j=1}^N e^{i\phi_j} \right|$.
- **Relational roughness:** $\mathcal{R} := \frac{1}{|E|} \sum_{\langle i,j \rangle} \langle (\phi_i - \phi_j)^2 \rangle$.
- **Bond-averaged relational distance:** $\overline{D} := \frac{1}{|E|} \sum_{\langle i,j \rangle} \frac{c}{\omega^*} \langle |\phi_i - \phi_j| \rangle$.

4.6 Helicity modulus (BKT diagnostic)

In simulation units $\beta = 1/T$:

$$\Upsilon = \frac{1}{N} \left\langle \sum_{(i,j)} J_{ij} \cos(\Delta\phi_{ij}) \right\rangle - \frac{\beta}{N} \left\langle \left(\sum_{(i,j)} J_{ij} \sin(\Delta\phi_{ij}) \right)^2 \right\rangle. \quad (8)$$

5 Transport ansatz (phenomenology with guardrails)

We introduce a first ansatz:

$$\sigma(T, \kappa) = \sigma_0 \frac{\Psi(T, \kappa)}{1 + \alpha \mathcal{R}(T, \kappa)}. \quad (9)$$

This is phenomenological. The factor $1/(1 + \alpha \mathcal{R})$ is the simplest saturating rational form at large $\mathcal{R}$.

5.1 Pre-registration discipline (pass/fail guardrails)

1. κ assigned **blind** to transport fits (Raman/AFM/WL proxies only).
2. J_0 fixed independently (e.g. reference low- T stiffness scale).
3. α is **global**: either fit on one reference dataset and predict all others (strictest), or fit once globally (declared before analysis).
4. λ and γ (if used) are **global** and fixed across samples.

5.2 Null models (required comparison)

$$\sigma \propto e^{-\alpha\mathcal{R}}, \quad \sigma \propto \frac{\Psi^2}{1 + \alpha\mathcal{R}}. \quad (10)$$

Model selection must favor Eq. (9) via objective criteria (AIC/BIC or cross-validated error), pre-registered.

6 Falsifiable predictions (kill tests)

6.1 Collapse variable

$$x := \frac{T}{J_0(1 - \kappa)}. \quad (11)$$

If extensions A/C are active, deviations from pure rescaling become *signal*, not nuisance.

Falsifiable prediction 1 (F1: collapse). *Across classes with independently assigned κ , the families $\Psi(x)$ and $\mathcal{R}(x)$ collapse to a master curve within finite-size uncertainty. Collapse quality is quantified by a pre-registered metric (e.g. reduced χ^2 with bootstrap confidence intervals) and compared against a fixed success threshold.*

Falsifiable prediction 2 (F2: Nelson–Kosterlitz universal jump). *At the BKT transition,*

$$\Upsilon(T_{BKT}^-) = \frac{2}{\pi} T_{BKT}, \quad (12)$$

in model units. Operationally, plot $\Upsilon(T)$ and $2T/\pi$; their intersection defines $T_{BKT}(L)$, then extrapolate using BKT finite-size forms (logarithmic corrections).

Falsifiable prediction 3 (F3: correlation exponent). *In the quasi-ordered phase,*

$$G(r) = \left\langle e^{i(\phi(r) - \phi(0))} \right\rangle \sim r^{-\eta}, \quad (13)$$

and at T_{BKT} one must observe $\eta \rightarrow 1/4$ within finite-size error bars, using an explicitly stated fit window $r_{\min} < r < r_{\max}$.

Falsifiable prediction 4 (F4: $\overline{D}(T_{BKT})$ constancy (optional)). *With ω^* fixed per class, $\overline{D}(T_{BKT})$ is approximately constant across κ in lattice units. Failure of F4 while F1–F3 hold falsifies the relational-distance layer while preserving the effective XY layer.*

7 Experimental mapping (data-first strategy)

7.1 Independent κ assignment (examples)

- **Raman defect proxy:** *D*-peak metrics (e.g. I_D/I_G) [2].
- **WL/WAL dephasing:** extract $\ell_\phi(T)$ from magnetoconductance fits [3, 4] and map $R \propto \ell_\phi^{-1}$.
- **Roughness:** AFM ripple amplitude (used only for ranking κ , not for fitting transport).

7.2 Priority dataset that is explicitly open (optical phase noise)

The phase-noise dataset associated with Ref. [1] provides direct access to time-series phase information suitable for extracting $\sqrt{\langle(\Delta\phi)^2\rangle}$ and comparing relational proxies across bandwidth choices.

7.3 Why graphene remains a natural first target

Graphene offers established WL/WAL phenomenology [3–5] and a mature transport context [6, 7]. When raw datasets are not openly posted, digitization of published curves or author requests can still support a first-pass test (declared explicitly in the analysis protocol).

8 Limitations (explicit)

- ϕ_i is an effective phase field (XY/BKT order-parameter style), not a fully microscopic electronic wavefunction phase.
- Eq. (9) is a first ansatz; acceptance requires objective superiority over null models.
- The framework does not claim “energy is unreal”: standard energetic quantities are treated as proxies for relational mismatch and phase-noise intensity, preserving experimental vocabulary.
- Extensions A/C are pre-registered: if data do not require them, the minimal model is preferred.

9 Quarantined programmatic layer (non-essential)

This manuscript is testable without invoking broader harmonic/binary-closure interpretations. Such interpretations are treated as a separate program and are not used to justify the falsifiers.

Companion materials (non-load-bearing)

Companion notes and prior drafts may be provided as related identifiers in repositories (e.g. Zenodo) but are not used as argumentative support in this manuscript.

A Workbench questions

1. Can κ be reduced to a composite index (Raman + WL/WAL + roughness) that remains device-agnostic?
2. Which minimal extension (vortex-core vs NNN coupling vs bath coupling $\Gamma(\kappa)$) best captures deviations from pure stiffness rescaling under fixed parameter count?
3. Under what conditions does $\overline{D}(T_{BKT})$ become a robust cross-platform marker, and when does it fail?

References

- [1] S. Donadello, E. K. Bertacco, D. Calonico, and C. Clivati. *Embedded digital phase noise analyzer for optical frequency metrology*. IEEE Transactions on Instrumentation and Measurement, 72: 1–10, 2023. doi: 10.1109/TIM.2023.3288255. Associated open dataset: <https://doi.org/10.5281/zenodo.8392622>.
- [2] L. G. Cançado, A. Jorio, E. H. Martins Ferreira, F. Stavale, C. A. Achete, R. B. Capaz, M. V. O. Moutinho, A. Lombardo, T. S. Kulmala, and A. C. Ferrari. *Quantifying defects in graphene via raman spectroscopy at different excitation energies*. Nano Letters, 11(8):3190–3196, 2011. doi: 10.1021/nl201432g.
- [3] E. McCann, K. Kechedzhi, V. I. Fal’ko, H. Suzuura, T. Ando, and B. L. Altshuler. *Weak-localization magnetoresistance and valley symmetry in graphene*. Physical Review Letters, 97: 146805, 2006. doi: 10.1103/PhysRevLett.97.146805.
- [4] F. V. Tikhonenko, A. A. Kozikov, A. K. Savchenko, and R. V. Gorbachev. *Transition between electron localization and antilocalization in graphene*. Physical Review Letters, 103:226801, 2009. doi: 10.1103/PhysRevLett.103.226801.
- [5] S. V. Morozov, K. S. Novoselov, M. I. Katsnelson, F. Schedin, L. A. Ponomarenko, D. Jiang, and A. K. Geim. *Strong suppression of weak localization in graphene*. Physical Review Letters, 97:016801, 2006. doi: 10.1103/PhysRevLett.97.016801.
- [6] A. H. Castro Neto, F. Guinea, N. M. R. Peres, K. S. Novoselov, and A. K. Geim. *The electronic properties of graphene*. Reviews of Modern Physics, 81:109, 2009. doi: 10.1103/RevModPhys.81.109.
- [7] S. Das Sarma, S. Adam, E. H. Hwang, and E. Rossi. *Electronic transport in two-dimensional graphene*. Reviews of Modern Physics, 83:407, 2011. doi: 10.1103/RevModPhys.83.407.